

Capstone Final Report

YouTube Trending Video Engagement and Reach (2024–2025)

Executive Summary

This project analyzed 200,000 sampled trending observations covering 91,033 unique videos across 113 countries from 2024-01-01 through 2025-12-31.

The typical trending entry earned a median likes-to-views ratio of 3.1%, while the top quartile began at about 5.3%.

Trending is fast: the median video first entered trending after 3 days, 85% did so within 7 days, and 97% did so within 14 days.

Friday-published videos showed the strongest median engagement in the sample, about 16% higher than Sunday-published videos.

A metadata-only Random Forest model achieved a ROC-AUC of 0.71 on the 2025 holdout set. Its top prediction decile contained a 44.9% high-engagement rate versus 4.4% in the bottom decile.

1. Business Problem

A client that manages YouTube content needs a practical way to understand what strong trending performance looks like, when videos typically surface in trending, and which content or publishing patterns are associated with stronger engagement.

The central challenge is that the dataset contains only videos that already trended. That means this project can explain patterns within the trending sample well, but it cannot estimate the true probability that any newly uploaded video will trend on YouTube overall.

2. Data and Sampling Approach

The source data is the Kaggle “Trending YouTube Videos — 113 Countries” dataset (<https://www.kaggle.com/datasets/asaniczka/trending-youtube-videos-113-countries/data>). It is a dataset of trending video observations across many countries.

First, the original file was read in chunks. Then the `snapshot_date` column was converted into a date format, so I could identify the year for each row. From this larger dataset, I kept only the rows where the year was 2024 or 2025, and then I sampled up to 100,000 rows from each year and saved the result as `youtube_2024_2025_balanced_sample.csv`. The file was a year-balanced analytical sample.

The sample contains 100,000 observations from 2024 and 100,000 from 2025. The project uses that 200,000-row file as the reproducible starting point. A fixed random seed of 42 was used in the sampling step so the sample could be reproduced from the raw file.

Because videos can appear multiple days and in multiple countries, the analysis used both row-level and video-level views of the data. Video-level analyses were used whenever repeated appearances would otherwise give too much weight to a small number of highly viral videos.

3. Analytical Approach

The workflow included four stages: data wrangling, exploratory data analysis, preprocessing, and modeling.

Feature engineering focused on publish timing, language, title and description structure, and text-style signals such as capitalization and punctuation. The key outcome metric for descriptive work was the likes-to-views ratio. The modeling target was whether a 2025 trending entry fell into the top quartile of likes-to-views ratio, with the quartile threshold defined from 2024.

4. Key Findings

Table 1 summarizes the most important descriptive findings that shaped the stakeholder narrative.

Metric	Result
Sample size	200,000 rows; 91,033 unique videos
Coverage	113 countries; 2024-01-01 to 2025-12-31
Median likes-to-views ratio	3.1%
Top quartile likes-to-views cutoff	5.3%
Median days to first trend	3 days
Trending within 7 days	85% of videos
Observed lower-tail first-entry views	10th percentile $\approx$ 106,928 views

A strict global “view threshold” for trending does not appear in the data. Instead, the lower tail varies substantially by country and context. In this sample, the 10th percentile at first appearance was roughly 107,000 views and the median was about 1.23 million views.

Finding 1: Most videos that trend do so quickly, which makes the first week after publication the most important operating window.

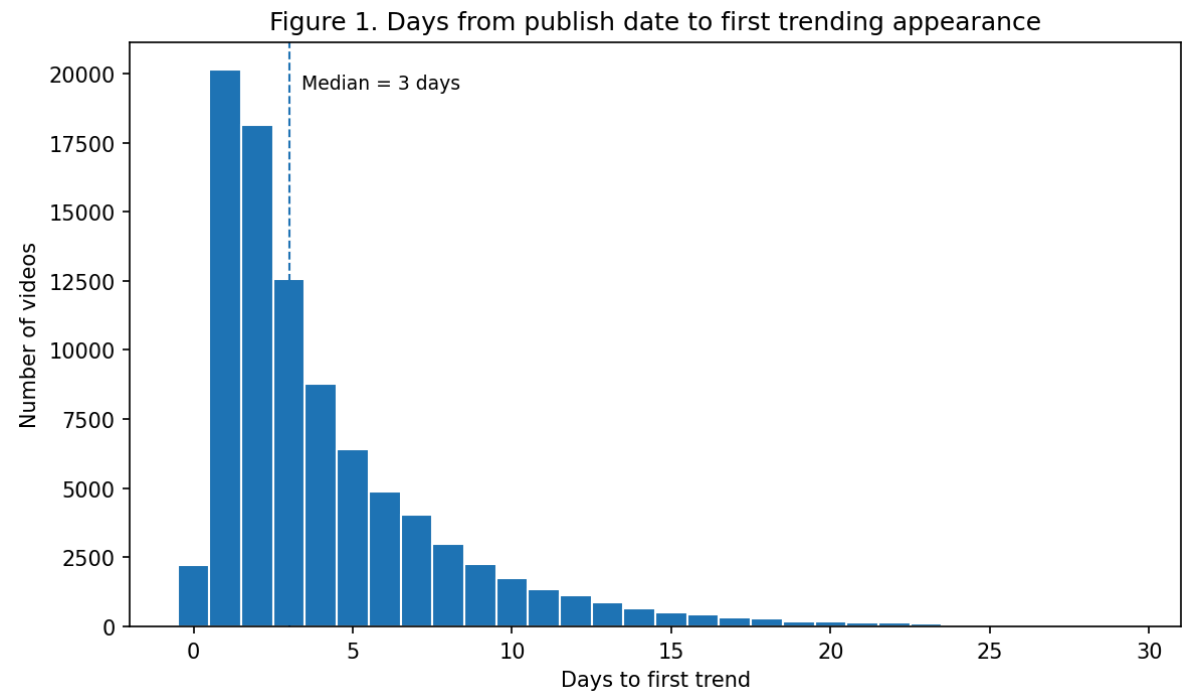


Figure 1. Distribution of days from publish date to first trending appearance.

The median time to first trending appearance was 3 days. More importantly for operations, 85% of videos first trended within 7 days and 97% within 14 days. This suggests creators and marketing teams should treat the initial release week as the critical period for amplification, community management, and creative iteration.

Finding 2: Publish timing matters within the trending sample, with Friday standing out as the strongest weekday for engagement.

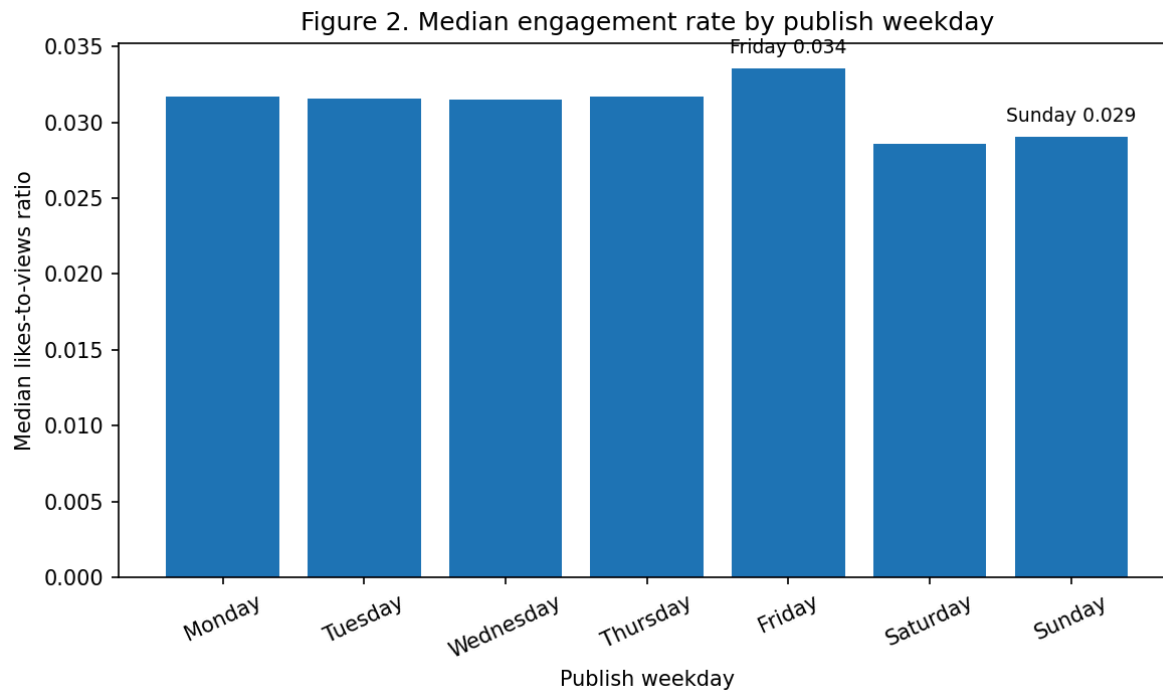


Figure 2. Median likes-to-views ratio by publish weekday.

Friday had the highest median engagement rate, while Sunday was one of the weakest. The Friday median was about 16% higher than Sunday in this sample. This should be treated as an association, not a causal guarantee, but it is strong enough to justify testing Thursday/Friday release calendars for content that is intended to travel quickly.

Finding 3: Language is associated with both cross-country spread and engagement.

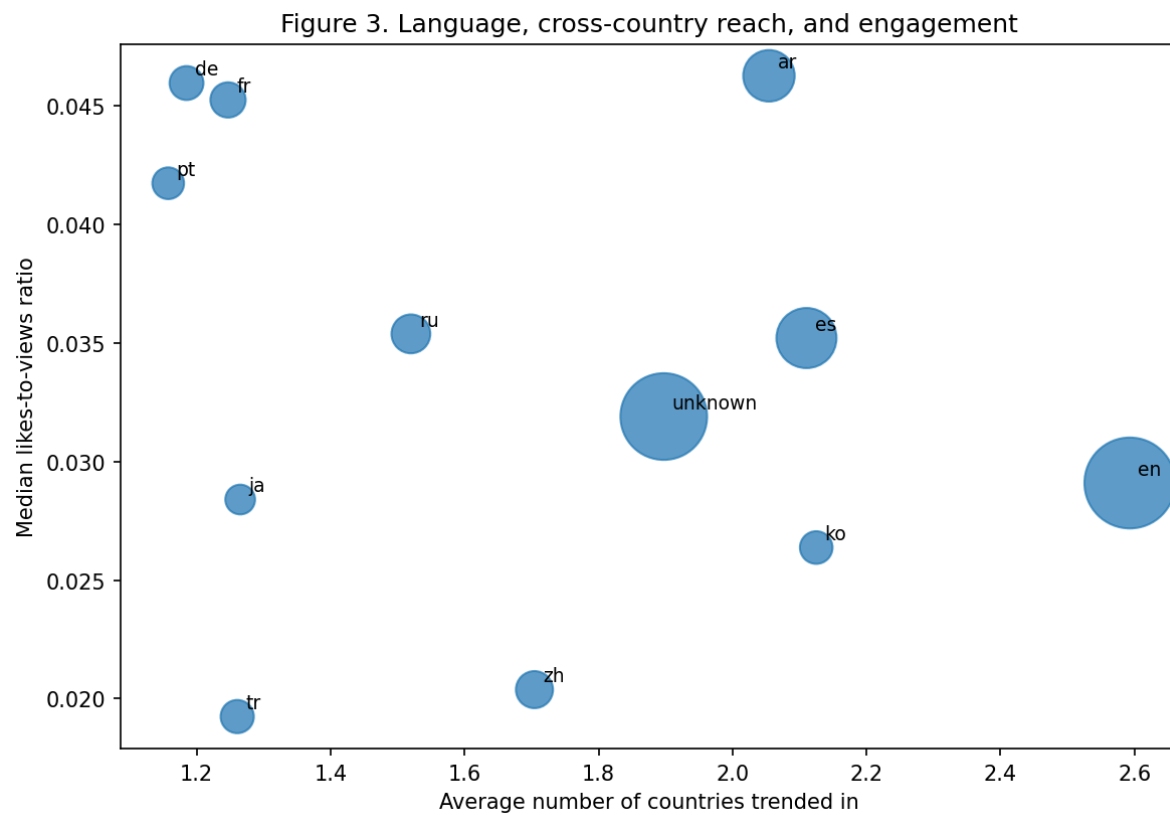


Figure 3. Major languages by average cross-country spread and median engagement.

English-language videos had the widest average cross-country spread in the sample, while Arabic and Spanish combined strong engagement with above-average multi-country reach. This suggests that localization decisions affect not only audience fit but also the potential breadth of spread across markets.

Additional metadata pattern: description length showed a non-linear relationship with engagement. Mid-length descriptions (roughly 301–600 characters) performed better than very short or very long descriptions on average. More emphatic title styles, especially titles with multiple exclamation points, also showed higher engagement in the sample. These findings are best interpreted as A/B testing candidates rather than rules.

5. Modeling Results

To add a predictive layer, the project modeled whether a trending entry in 2025 would land in the top quartile of likes-to-views ratio. The model used metadata and publish-context features rather than direct engagement counters like likes or comments, to avoid target leakage.

Model	ROC-AUC	Avg Precision	Accuracy	Precision	Recall	F1
Logistic Regression	0.700	0.366	0.605	0.294	0.707	0.415
Random Forest	0.713	0.379	0.635	0.311	0.692	0.429

The Random Forest model performed best, with a ROC-AUC of 0.71. That is useful but not production-grade certainty. In practice, the model is best used as a ranking and screening tool, not as a final decision-maker.

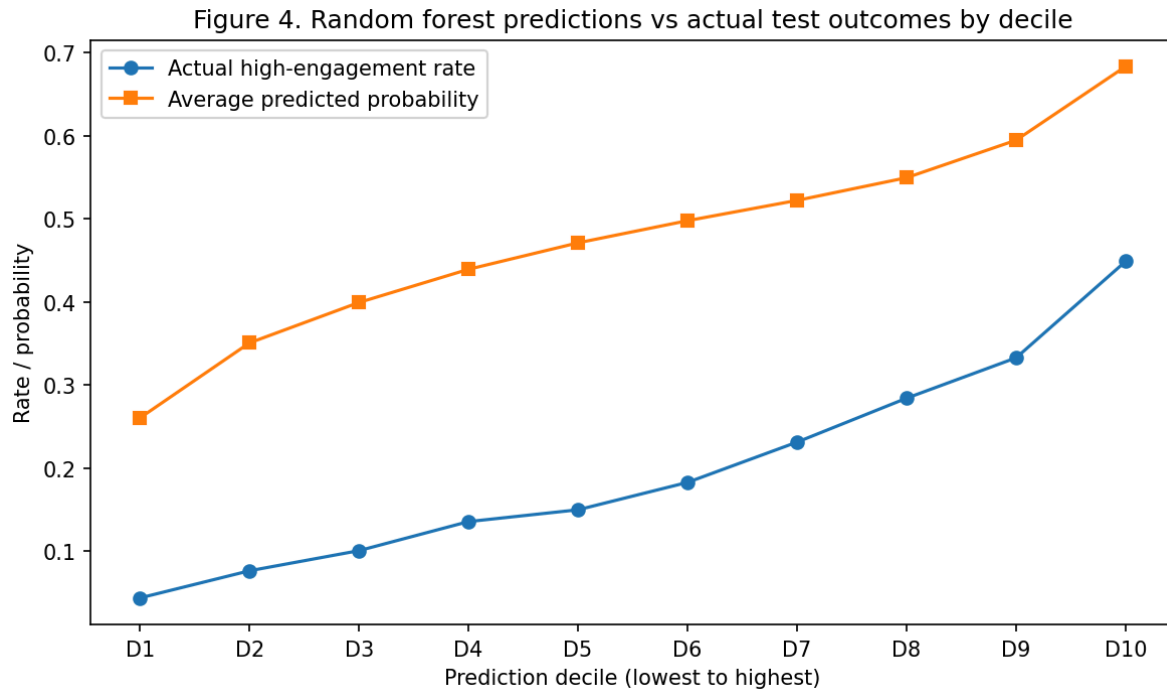


Figure 4. Random Forest predictions compared with actual high-engagement outcomes in the 2025 test set.

The top prediction decile contained a 44.9% high-engagement rate, versus 4.4% in the bottom decile. This shows the model can separate stronger from weaker cases even though classification error remains substantial at the individual-video level.

6. Recommendations for the Client

1. **Prioritize the first-week operating window.** Plan promotion, creator follow-up, community moderation, and thumbnail/title testing around the first 7 days after release. Most videos that eventually trend enter the list very quickly.
2. **Test Thursday/Friday release scheduling for priority launches.** Friday-published videos showed the strongest median engagement in this sample. For key releases, test late-week launch timing against current norms instead of assuming all weekdays perform equally.
3. **Use localization strategically for cross-market content.** English offered the broadest spread, while Arabic and Spanish combined strong engagement with multi-country reach. For content with export potential, test localized metadata, captions, and packaging in these language groups.

7. Limitations

The dataset contains only trending videos, so results describe behavior within the trending sample rather than the entire universe of YouTube uploads.

Language labels include an “unknown” bucket and may contain some noisy detections. Country-level trending mechanics also likely differ in ways the current model does not explicitly estimate.

8. Ideas for Further Research

Add a richer text model for titles and descriptions, including sentiment, entities, and topic embeddings.

Model country-specific behavior separately to test whether view-count lower tails or publish-day effects differ by market.

Track repeated appearances over time to estimate retention on the trending list, not just first entry and engagement level.